9. The following table gives the number of points scored by a rugby team in a season.

Points scored	0–4	5–9	10–14	15–19	20–24
No. of matches	3	9	15	10	3

(i) How many matches did the team play that season?
(ii) Use mid-interval values to calculate an estimate of the mean number of points scored, correct to the nearest whole number.
(iii) What was the modal class of this data?
(iv) What was the least number of points, greater than the mean value, scored?

10. The test results of 51 students were grouped as follows:

Mark	30–39	40–49	50–59	60–69	70–79
Frequency	7	13	22	8	1

Find the (i) mean (ii) mode (iii) median of the data in the table.

Section 7G The Range

When comparing sets of data it is important to focus on two things:

(i) An average (mean, mode or median), a value that gives a sense of the centre or middle of the data.
(ii) The range which is a measure of how spread out the data is.

Range The range for a set of data is the largest value of the set minus the smallest value of the set.
Range = (Largest − Smallest) value

Example 1

The marks out of 20, scored by Ava and Sean in 6 maths tests are given below:

Ava	12	10	18	14	8	10
Sean	12	10	16	13	11	10

Compare Ava's and Sean's results.

(i) Comparing means: Ava: $\frac{12 + 10 + 18 + 14 + 8 + 10}{6} = 12$

Sean: $\frac{12 + 10 + 16 + 13 + 11 + 10}{6} = 12$

(ii) Comparing ranges: Ava: $18 - 8 = 10$

Sean: $16 - 10 = 6$

Although they both have the same mean value, Ava's results are more spread out.

Exercise 7G

1. Find the range of each of the following sets of data.
(i) 3, 7, 9, 18, 21, 28
(ii) 8, 2, 9, 6, 10, 11, 1, 18, 3, 23
(iii) 3.1, 4.5, 2.8, 6.1, 4.3, 9.1, 1.7, 5.4

2. The range of the set of data 2, 4, 5, 9, x is 14. What is the value of x?

3. The times for a bus journey in minutes are 15, 7, 9, 12, 9, 19, 6, 11, 9, 16, 8.

(i) Find the range of these times.

(ii) Find the mode and median of these times.

4. A greengrocer sold boxes of apples from different countries.
A box contained 9 French apples.

The weight of each apple recorded in grams is given below:
101, 107, 98, 109, 115, 103, 96, 112, 104

(i) Calculate the mean weight of a French apple.

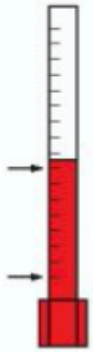
(ii) Find the range of weights of the French apples.

Another box contained 9 South African apples.

Their mean weight was 107 g and their range was 19 g.

(iii) Make two comments about the weights of the apples from the two boxes.

5. The temperature in a glasshouse was measured over 7 days at the same time of day.
The data collected were:
4°C, 3°C, 3°C, −1°C, 0°C, 2°C, −2°C



(i) Find the range of temperatures over the 7 days

(ii) Find (a) the mean and
(b) the median value of the temperature.

6. (i) Write 5 numbers that have a range of 6

(ii) Write 5 numbers that have a mode of 3 and a range of 4

(iii) Write 5 numbers that have a median value of 7 and a range of 10.